Privacy-Preserving Multi-Agent Intelligent Chatbot for PCOS Diagnosis

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Outline

1 Motivation

2 Proposed System

3 Controllable History-Taking Agent

4 Status and Future Work

5 Conclusions

Polycystic Ovary Syndrome (PCOS)

- One of the most prevalent endocrine disorders in women of reproductive age.
- Dysregulation of the hypothalamic–pituitary–ovarian (HPO) axis:
  - disrupted GnRH secretion $\Rightarrow$ elevated LH, suppressed FSH;
  - insulin acts with LH to drive excess ovarian androgen production;
  - the hyperandrogenic environment disrupts follicle maturation $\Rightarrow$ chronic anovulation.
- Left undiagnosed, PCOS raises long-term risks of type 2 diabetes, cardiovascular disease, endometrial hyperplasia, infertility, and psychological comorbidity.
- No definitive cure, but early diagnosis + lifestyle changes limit metabolic complications at low cost.

A Growing, Under-Diagnosed Public Health Challenge

10–13%
of women of reproductive age are affected
(WHO)

Up to 70%
of affected women are unaware of their condition
(WHO)

+88.95%
prevalent cases, women aged 15–49, 1990→2021
(34.8M → 65.8M, *GBD 2021*)

33.6%
waited more than two years for a diagnosis
(international survey, $n = 1{,}385$)

47.1%
consulted three or more health professionals before diagnosis

91%
of women with PCOS are willing to use a dedicated app

Delays reflect a knowledge gap: patients lack the information needed for self-management.

Why Patients Hold Back

- Smart healthcare systems support history-taking, decision support, diagnosis, and monitoring.
- But a consultation only works if patients share relevant information.
- Reproductive history is intimate: menstrual, sexual, reproductive.
  - Disclosure is held back by embarrassment, stigma, fear of judgment, and privacy concerns.
  - A digital agent does not by itself increase disclosure.
- What matters: trust in the system, transparency about data use, and perceived privacy and security.

Design objective

Facilitate *appropriate* disclosure, not just collect more data, while keeping sensitive health information private, secure, and confidential.

Explainability: Prerequisite for Trust and Compliance

- "Black-box" models limit trust and clinical adoption, and raise bias and compliance concerns.
- XAI lets users validate AI-assisted decisions instead of accepting them blindly.
- Supports biomedical ethics: beneficence, non-maleficence, autonomy, justice.
- GDPR, HIPAA, and the EU AI Act demand transparency and human oversight, but do not prescribe how explainability is implemented.

Transparency

Show how and why an output was produced.

Accountability

Trace every decision to a specific model version.

Fairness

Mitigate bias towards particular patient subgroups.

Our Approach

A privacy-preserving, multi-agent intelligent chatbot for early PCOS screening and education, combining in one auditable interface:

Multi-agent

Specialised agents for history taking, test preparation, and diagnosis support.

Explainable AI

Predictions come with the reasoning clinicians need to check them.

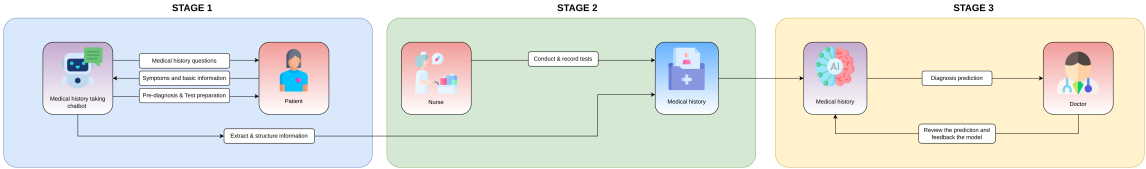
Retrieval-augmented generation

Answers grounded in trusted medical sources.

Privacy by design

Privacy-enhancing techniques and secure access to sensitive data.

System Architecture



Three stages from first contact to diagnosis, with a clinician in the loop at every step.

The Three Stages

1 History taking

Chatbot & patient

- Chatbot asks medical history questions.
- Patient shares symptoms and basic information.
- Chatbot gives pre-diagnosis and test preparation.
- Answers are extracted and structured into the medical history.

2 Clinical testing

Nurse

- Nurse conducts the tests the patient was prepared for.
- Results are recorded in the same structured medical history.

3 Diagnosis support

AI model & doctor

- Model predicts a diagnosis from the complete medical history.
- Doctor reviews the prediction.
- Doctor's feedback improves the model.

Research Questions

History-taking as a constrained POMDP: policy π, extractor f, quality Q

RQ1 – Policy π

How should the agent choose its next action?

- Cover the PCOS guideline fields
- Follow the communication protocol
- Respect consent and safety

RQ2 – Extractor f and quality Q

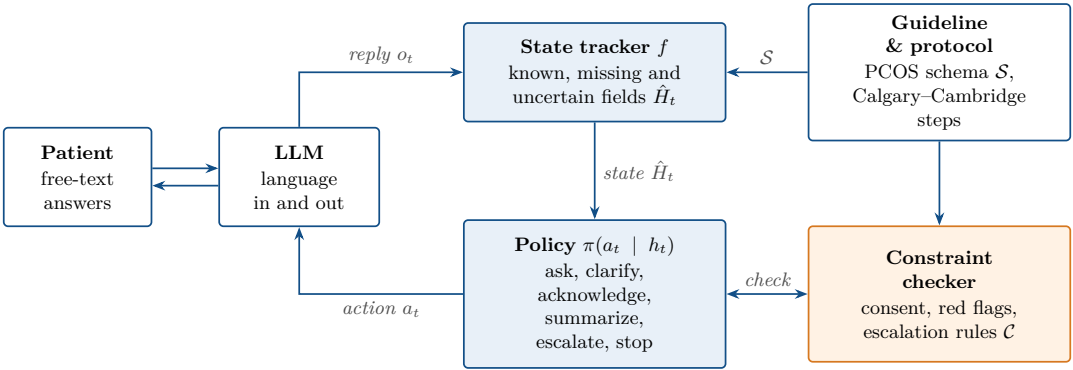
How do we extract and assess a patient history?

- Facts, timing and chronology
- Evidence and uncertainty per field
- Coverage and factual accuracy

Initial use case: PCOS. Other conditions need their own schema and protocol.

Proposed Agent Architecture

The LLM handles language; state, policy and constraints stay explicit



Evaluation Plan

Planned evaluation; no results are reported yet

Data and setting

- MediTOD dialogues for extraction
- Simulated PCOS patients built from guideline profiles z
- Clinician-reviewed pilot cases

Metrics

- **RQ2:** field coverage, accuracy, temporal consistency, unsupported claims
- **RQ1:** protocol adherence, turns T, correct escalation

Baselines

- Prompt-only LLM
- Rule-based guideline checklist
- Hybrid: LLM with explicit policy and constraints

Reference standard: clinician-reviewed histories.

Project Resources

- **Video presentation** of the system: coming soon.
- **Interactive demo** of the chatbot: coming soon.
- **Project page**: `https://innovher-competition.github.io/demo/`

Conclusions and Next Steps

Key messages

- For PCOS, the interview can inform two of three diagnostic criteria
- History-taking formulated as a constrained POMDP: policy π, extractor f, quality Q
- The proposed agent keeps state, policy and constraints explicit around the LLM

Next steps

- Build the PCOS schema $\mathcal{S}$ from the 2023 guideline
- Implement prompt-only, rule-based and hybrid policies
- Evaluate on MediTOD and simulated patients, then with clinician review

Scope

Clinician-reviewed history collection, not autonomous diagnosis.

Thank you

Questions?

`https://innovher-competition.github.io/demo/`

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